1 **(a)** Continuous **[B1]**

(b) Discrete **[B1]**

(c) Continuous **[B1]**

[Total Marks: 3]

2 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(iii) Continuous **[B1]**

(iv) Discrete **[B1]**

[Total Marks: 4]

3 **(a)** Discrete **[B1]**

(b) Continuous **[B1]**

(c) Continuous **[B1]**

(d) Continuous **[B1]**

[Total Marks: 4]

4 **(a)** Discrete **[B1]**

(b) Continuous **[B1]**

(c) Continuous **[B1]**

(d) Discrete **[B1]**

[Total Marks: 4]

5 **(a)** Data Set A (time) is continuous, so it can take any decimal value and inequalities are needed to define exact boundary limits without gaps. **[B1]**

Data Set B (number of push-ups) is discrete, so it only takes integer values and a hyphen clearly defines the inclusive start and end integers. **[B1]**

(b) **(i)** 19.5 and 29.5 **[B1]**

(ii) 10 **[B1]**

(c) Age (usually given as an integer in years) OR Money/Currency (rounded to nearest cent/pence) OR Shoe size **[B1]**

[Total Marks: 5]

6 **(a)** Continuous **[B1]**

(b) Height is continuous / can take decimal values. **[B1]**

A plant with a height between the intervals (e.g. 19.5 cm) would not fit into any class. **[B1]**

(c) Discrete **[B1]**

[Total Marks: 4]

7 (a) Discrete data can only take specific or exact values (often counted) **[B1]**

Continuous data can take any value within a given range (often measured) **[B1]**

(b) (i) Continuous **[B1]**

(ii) Rounding limits the precision of the recording, but the actual true length could still be any value between the rounded boundaries **[B1]**

[Total Marks: 4]

8 (a) Continuous **[B1]**

(b) Uses midpoints: 90, 110, 130, 150 **[M1]**

$\Sigma fx = 90 \times 12 + 110 \times 26 + 130 \times 15 + 150 \times 7 = 6940$ **[M1]**

$\frac{6940}{60} = 115.66... = 116$ or 115.7 **[A1]**

(c) The exact masses of the apples in each class are not known (or similar reference to data being grouped) **[B1]**

[Total Marks: 5]

9 (a) Length **[B1]**

(b) (i) 14.45 **[B1]**

(ii) 14.55 **[B1]**

(c) Identifies that the discrete values include $5, 6, 7, 8, 9$ OR uses continuous boundaries $9.5 - 4.5$ **[M1]**

5 **[A1]**

[Total Marks: 5]

10 (a) (i) Discrete **[B1]**

(ii) Continuous **[B1]**

(b) The variable can still only take specific, fixed intervals/values (e.g., jumping by 0.5) **[B1]**

It cannot take any value within a range (e.g., a shoe size of 8.23 does not exist) **[B1]**

[Total Marks: 4]

11 **(a)** Continuous data can take any value within a given range (measured). **[B1]**

Discrete data can only take specific, exact values (usually counted). **[B1]**

(b) **(i)** Continuous **[B1]**

(ii) Shoe size can only take specific, set values (there are gaps between sizes like 7 and 7.5). **[B1]**

It cannot take every possible value (e.g. you cannot have a shoe size of 7.23), so it must be discrete. **[B1]**

[Total Marks: 5]

12 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(b) Upper bound of mass of one book = 855 g **[M1]**

$24 \times 855 = 20520$ g $\implies$ 20.52 **[A1]**

(c) Lower bound of empty box = 1.15 kg **[B1]**

Lower bound of mass of one book = 845 g = 0.845 kg seen **[B1]**

Total lower bound $= 1.15 + (24 \times 0.845) = 21.43$ **[A1]**

[Total Marks: 7]